Classic hedging is all you need

Classical versus learned option hedging under transaction costs

Sepp (2013) and Buehler et al. (2019) replications, plus a head-to-head comparison

DSC4107 Final Project

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The problem

- Sell an at-the-money one-month call, hedge it with the underlying.
- Trades happen at discrete times, and each one costs a fixed fraction of notional.
- More trading tracks the delta but pays more cost; less trading saves cost but leaves variance.

Two papers attack the same trade-off from opposite ends:

Sepp (2013)
Closed-form P&L statistics for delta hedging. Maximises the Sharpe ratio over rebalancing frequency, then converts it to price and delta bands.

Buehler et al. (2019)
A neural network chooses the hedge from market state, trained on a convex risk measure of terminal P&L.

Neither paper runs the two against each other under controlled conditions. That comparison is our contribution.

Research questions

- Can a Monte Carlo simulator reproduce Sepp's closed-form statistics, tables and figures?
- Can we reproduce the main claim of Deep Hedging, that a trained network beats Black-Scholes delta hedging under transaction costs in a Heston market?
- Once the classical side is re-tuned per cell, does the learned hedger beat the best band at 20 bp—and where does the ranking change with cost?

Headline answer (RQ3): at 20 bp, classic hedging is enough. High-cost Sharpe wins are premium harvesting, not better hedges.

Methods

Testbed

- Four market models: GBM, jump-diffusion, Heston, Heston with jumps
- Sepp's parameters, so replication lines up
- One ATM one-month call, one proportional cost, one shared P&L rollout
- Cost charged on opening and closing trades too

Comparison design

- Classical baseline re-tuned in every cell
- Learned network: moneyness, time, current position; CVaR loss
- Disjoint train, validation, tuning, search and holdout seed pools
- Every number averages three training seeds

Correctness anchor. With zero cost the learned policy must recover the Black-Scholes delta (correlation 0.9907).

Replication 1: Sepp (2013)

- Largest relative error in `u`: 3.9%
- GBM and SVJ case I match to 0.0001
- 60,000-path Monte Carlo agrees on mean and volatility

Jump-diffusion fix: restore factor `q = sqrt(pi^3)/4`; analytic JD Sharpe matches simulation (0.65).

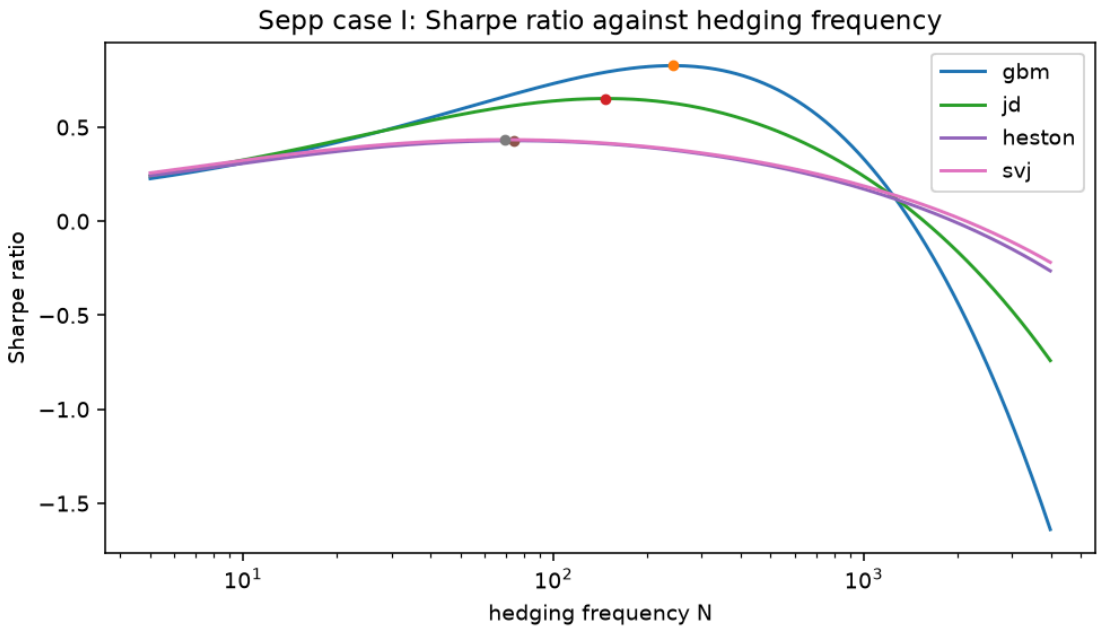


Figure 1: Sharpe against hedging frequency, case I.

Replication 2: Deep Hedging

- Frictionless check validates the implementation
- Heston: matches delta at zero cost; beats every-step delta as cost rises
- Qualitative claim of the paper is reproduced

This is not the headline: the frozen 20 bp benchmark is Section 7.

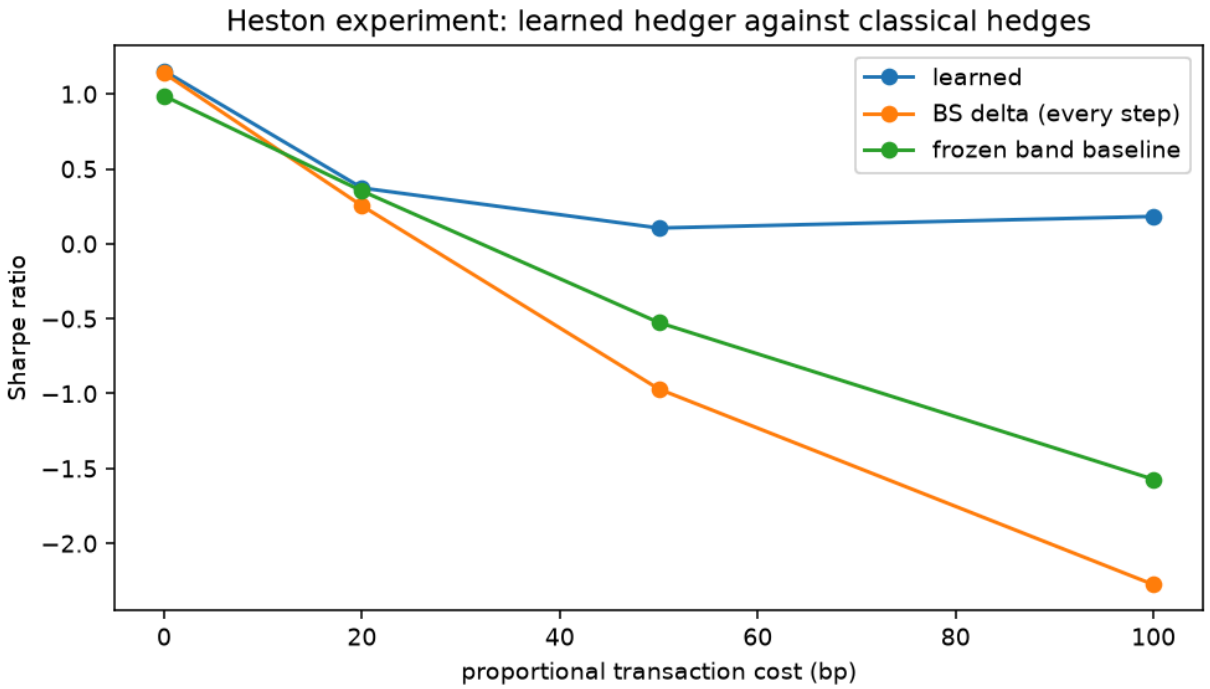


Figure 2: Heston Sharpe against cost.

20 bp: classical vs deep

- **Near zero cost:** tie (frictionless check)
- **At 20 bp:** Sharpe edges −0.09 to +0.01 — essentially flat
- Jump models: classical slightly ahead
- Holdout: mean edge +0.0339 — real but tiny

Classic hedging is all you need at nominal friction.

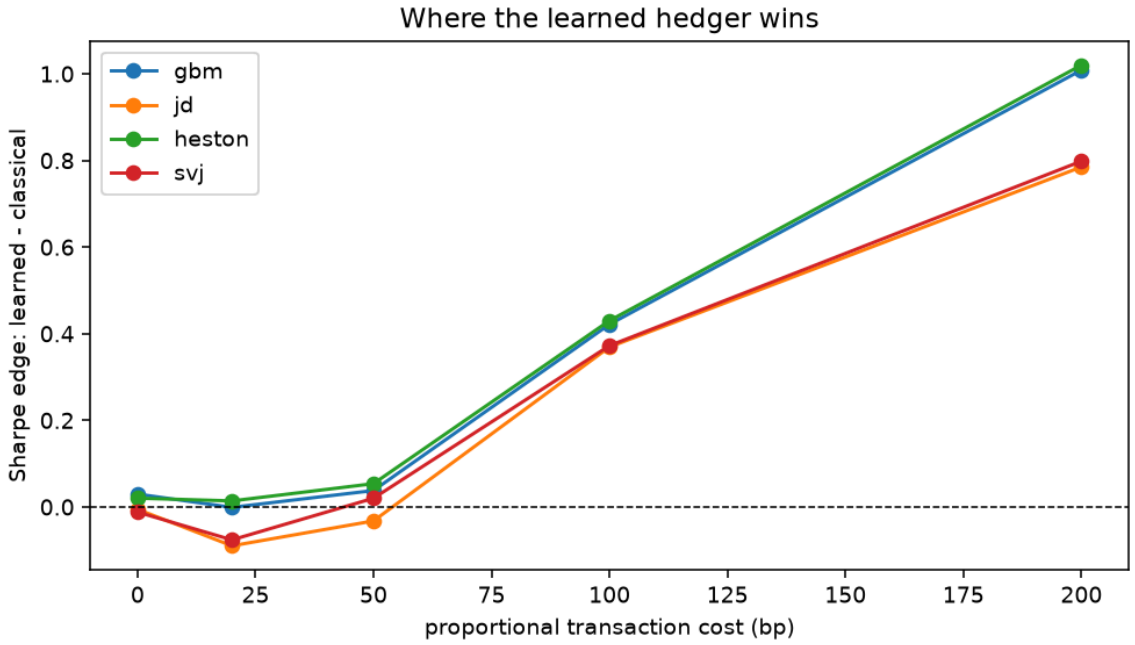


Figure 3: Learned minus classical Sharpe (fixed hyperparameters).

Same spread, worse tail — and high-cost "wins"

- Learned vs band: mean 0.240 vs 0.216, std 0.626 vs 0.613
- Skew −0.70 vs −0.01; CVaR 95% 24% worse
- Sharpe edge buys mean with a fatter left tail
- **Above 50 bp:** large Sharpe edge but std rises to 3–4
- P&L mean approaches the option premium → premium harvesting

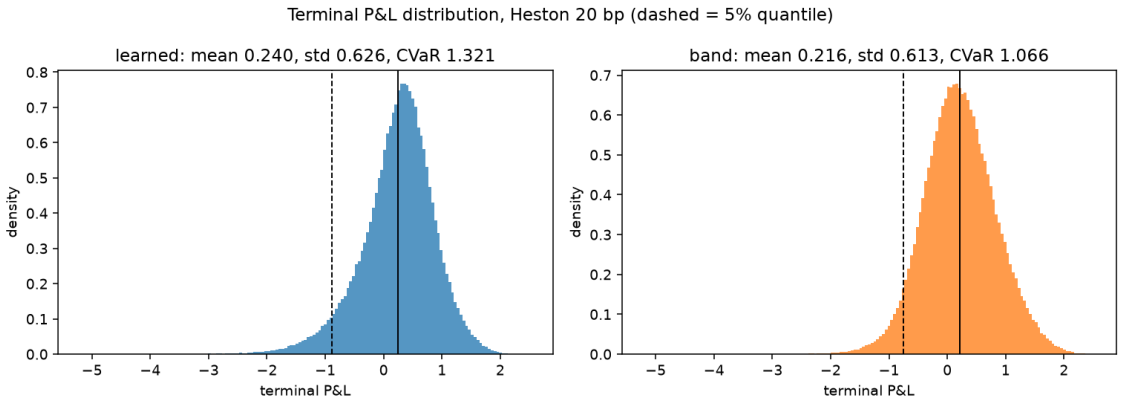


Figure 4: Terminal P&L on the frozen benchmark.

Liquid 10 bp (footnote only)

- Sepp Case I charges **k/2** on $|\Delta\delta| \cdot S$ (eq. 2.5); side study at liquid-index **k** uses our **full-rate** ledger (`cost = 0.001` — numerically Sepp's **k**, not Case I economics)
- 10 bp classical hull re-swept; not `classical_best.json`
- H `clears` (narrow) → K `dominated` → L `a05_only` → I `fails` → J `band_like`

Recipe-specific, mandate-dependent, brittle. Does not reopen the 20 bp headline.

Conclusion

- Replications hold: Sepp Table 2, Monte Carlo cross-check, frictionless delta recovery.
- At 20 bp a re-tuned classical band does not lose on Sharpe; deep is not a uniform upgrade.
- High-cost Sharpe gains track under-hedging and premium collection — report them with P&L spread and tail risk.

github.com/hluaguo/dsc4107

Full tables: `submission/report/report.pdf` and root `REPORT.md`.